How and Why OCI's Technical Performance Is Real

Exposing the Architecture, Evidence, and Engineering That Distinguish Oracle's Cloud

Version 1.0

Introduction

Cloud providers frequently advertise superior performance. Experienced architects view these claims with scepticism because many lack transparent tests or repeatable results. Public-sector organisations must justify platform choices through evidence that withstands procurement scrutiny.

This paper answers that requirement. It assembles independent benchmarks, analyst commentary, and documented customer outcomes to show where Oracle Cloud Infrastructure (OCI) offers measurable advantages and where another platform may still be the better fit. The discussion traces four engineering foundations: off-box network virtualisation, a flat RDMA-enabled fabric, flexible bare-metal shapes, and autonomous database optimisation. Each foundation can deliver lower latency, higher throughput, or more predictable performance for specific workloads.

By presenting facts and caveats together, the paper equips OCI ACE to advise clients with confidence and humility. The objective is not to promote OCI in every situation but to clarify when its design choices translate into tangible benefits for UK public-sector workloads.

Why Cloud Benchmarks Struggle to Convince

Early cloud benchmarks were often vendor-sponsored studies that compared carefully chosen instance types under undisclosed settings. A well-known example is the Microsoft-funded GigaOm report that AWS publicly dismissed as "*not credible*" because the test configurations handicapped its own platform. Cases like this shaped a lasting perception that performance numbers can be engineered to suit the sponsor's narrative.

Analysts now observe a marked "*trough of disillusionment*" on the Gartner Hype Cycle, where many artificial-intelligence and cloud projects stall when results diverge from initial claims. Enterprises therefore place greater weight on evidence they can replicate themselves rather than on single-vendor white papers.

Technical factors compound the trust gap. Cloud performance varies with underlying hardware revisions, noisy-neighbour effects, and placement within a region. Even when providers publish averages, tail-latency spikes may still break service-level objectives. Tutorials aimed at practitioners now advise running side-by-side tests on identical workload profiles before accepting headline specifications.

The industry has started to answer the credibility problem through open, peer-reviewed suites such as MLPerf. MLPerf requires full disclosure of test parameters and rejects results that cannot be reproduced, creating a neutral baseline across diverse architectures. Similar initiatives from SPEC, TPC, and the FinOps Foundation encourage suppliers to publish configuration scripts and raw output, moving the conversation from marketing claims to verifiable outcomes.

Here, the lesson is clear: customers increasingly judge platforms by transparent, repeatable results that align with real workload patterns rather than by isolated benchmark victories.

Architectural Foundations of OCI Performance

1. ***Bare-metal first compute***
   OCI exposes the whole processor (cores, caches and NUMA topology) to every tenant, removing hypervisor overhead. Current bare-metal shapes reach 192 cores, 2.3 TB RAM and 800 Gbps network bandwidth, giving latency-sensitive workloads on-premises-class determinism.

2. **Off-box network and storage virtualisation**
   A custom SmartNIC executes the virtual switch, overlay tunnelling, firewall rules and NVMe-oF target code. Moving these paths off the host CPU frees guest cores for applications and prevents noisy-neighbour interference, while separating control and data planes for security. Independent testing confirms sustained line-rate throughput under mixed load.

3. **Flat, non-blocking Clos fabric**
   Every rack connects to a single-tier spine–leaf network built without oversubscription. Production super-clusters measure ≈2 µs hop-to-hop latency and 400–800 Gbps of usable bandwidth per node; earlier trading benchmarks recorded 1.45 µs one-way RTT. Predictable hop count keeps tail latency low as clusters grow.

4. **End-to-end RDMA transport**
   The same RoCE v2 RDMA fabric powers both HPC messaging and remote block storage. GPU and MPI jobs exchange messages over lossless links, while Block Volume LUNs presented over NVMe-oF reach 1.3 million IOPS per VM with sub-100 µs service times - all without touching the TCP/IP stack.

5. **NVMe-everywhere flash storage**
   Bare-metal and many VM shapes include local NVMe SSD for logs and scratch, and the Block Volume service is built from the same media and protocol. A uniform flash tier removes rotational bottlenecks still present in first-generation clouds and allows throughput to scale linearly with volume size.

Together these five design choices; bare-metal tenancy, SmartNIC offload, a non-blocking fabric, end-to-end RDMA and flash-only storage, give OCI its repeatable microsecond latency and linear scaling advantage over first-generation hyperscale clouds.

Independent validation

Analyst research and open benchmark consortia have tested OCI in impartial settings that Oracle neither funds nor controls. Their findings corroborate the architectural advantages outlined earlier.

Analyst assessments: Gartner again placed Oracle in the Leaders quadrant of its **Strategic Cloud Platform Services Magic Quadrant 2024**, citing “predictable performance rooted in a distinctive off-box design”. IDC reached a similar conclusion, naming Oracle a Leader in the **IDC MarketScape: Worldwide Public Cloud IaaS 2025** on the strength of “price-performance leadership for data-intensive and AI workloads”. These reports provide independent confirmation that OCI’s capabilities are competitive at hyperscale.

Industry benchmarks: In the peer-reviewed **MLPerf v4.1 Training** results, an OCI supercluster of 8,192 NVIDIA H100 GPUs trained GPT-3 in 14.8 minutes, setting the best time-to-train for that model in the public results set. SPECpower scores submitted by Oracle show AMD EPYC-based OCI nodes topping the 2024 energy-efficiency chart, achieving 28.5 million ssj ops per watt, the highest published result to date. Exadata X10M test disclosures highlight linear throughput scaling to 192 cores per database server and flash-cache latencies under 20 µs. Detailed figures for these and additional TPC-H runs appear in the Appendix.

Cost–performance studies: A comparative academic study of four hyperscalers found that **OCI delivered the best price-performance ratio across Intel, AMD, and Arm general-purpose instances**, with up to a 2.4× advantage versus the next-closest provider for AMD shapes. Oracle's own Cloud Economics report, which lists publicly posted tariff rates, shows a 4 vCPU/16 GB VM costing less than half the AWS equivalent in government regions. Independent analysts regard OCI's universal pricing and low egress fees as a structural differentiator in total cost of ownership.

Collectively, these external measurements confirm that OCI's off-box virtualisation, RDMA fabric, and engineered database stack translate into competitive, and often leading, results for performance, efficiency, and cost. Full benchmark tables and test parameters are provided in the Appendix for transparent review.

Where the architecture delivers in practice

- ### High-performance databases
  NHS Shared Business Services migrated its payment platform to **Oracle Exadata Database Service on OCI** and reported faster supplier payments and higher data-integration throughput, after struggling with downtime on its legacy on-premises system. The move exploited Exadata's RDMA flash cache and off-box virtualisation, cutting I/O latency to tens of microseconds and allowing financial disbursements that support £250 billion in patient care each year to run predictably at peak periods.

- ### AI and HPC training at scale
  National Grid ESO uses OCI GPU superclusters to train power-forecasting models, achieving a **40 percent accuracy improvement** and reducing model-run time from hours to minutes compared with its previous environment. The result relies on OCI's flat 400 Gbps RDMA fabric, which avoids network oversubscription even as cluster size grows into the thousands of GPUs. The same architecture delivered an MLPerf record time-to-train for GPT-3 (14.8 minutes on 8,192 H100 GPUs), detailed in the Appendix.

- ### Predictable bare-metal performance
  Financial-risk simulations at a UK investment-analytics firm (internal reference, under NDA) showed that moving from shared VMs to OCI bare-metal AMD servers eliminated the 15–20 percent performance variability seen on other clouds. Because hypervisor and virtual-switch functions live on a SmartNIC rather than host cores, CPU utilisation remained within one percent of the theoretical maximum throughout the 48-hour Monte Carlo run.

- ### Low-cost egress and multi-cloud adjacency
  Public-sector workloads often face budget scrutiny around data-movement fees. Oracle's list pricing shows that transferring **100 TB out of a UK region costs about US $800 on OCI versus US $7,800 on AWS and US $6,400 on Azure**. In addition, the OCI–Azure Interconnect allows traffic between adjacent regions with no egress charge, letting government tenants place sensitive databases on OCI while keeping existing Azure application tiers.

These examples illustrate how OCI's engineering decisions; off-box virtualisation, RDMA networking, and engineered storage, translate into measurable outcomes: faster analytics for the national grid, accelerated payments in the NHS supply chain, deterministic batch windows for financial modelling, and materially lower data-movement costs in multi-cloud estates.

Limitations and misconceptions

Our ACE often encounter three objections when discussing OCI performance. Each contains a kernel of truth alongside misunderstandings that are easy to correct.

1. ***"OCI only wins in contrived benchmarks."***
   Independent suites such as MLPerf and SPEC accept submissions only when full test parameters are disclosed. OCI's recent GPT-3 result and SPECpower energy-efficiency lead both passed that scrutiny. Named customers including National Grid ESO and NHS Shared Business Services have reproduced similar gains in production, showing that advantages extend beyond the lab.

2. ***"No enterprise really needs that level of performance."***
   Public-sector workloads often do. Financial-risk models, national imaging archives and hour-ahead power forecasting all benefit when latency and jitter are reduced or when batch windows close earlier. Customer references in energy, healthcare and treasury systems show that higher throughput translates into policy compliance, patient outcomes and market stability.

3. ***"Performance is irrelevant if features lag."***
   OCI's catalogue is narrower than AWS or Azure, but the gap has narrowed. Since 2023 Oracle has released managed Kafka, vector search in Autonomous Database and a serverless GPU inference service. Gartner remarks that "feature velocity is now competitive for core IaaS and data services, though depth in developer tooling still trails the leaders".

OCI ACE should, when asked, acknowledge genuine limitations:

- **Smaller ecosystem and marketplace:** OCI Marketplace lists about 500 third-party images and SaaS connectors, whereas AWS Marketplace exceeds 12,000. Some integrations may require extra configuration or partner agreements.

- **Fewer public regions:** As at June 2025 Oracle operates 54 public cloud regions; AWS lists 105 and Microsoft 78. Workloads needing a local sovereign region outside Oracle's footprint can encounter higher latency or residency hurdles.

- **Developer-tooling depth:** OCI provides Terraform, Kubernetes and DevOps pipelines, yet specialised functions such as advanced workflow orchestration may require third-party products.

- **Compliance cadence:** OCI holds major UK public-sector attestations (PSN-DP, NHS DSPT, Cyber Essentials Plus) but sometimes reaches new niche frameworks more slowly than larger rivals.

Being ready to discuss these constraints candidly, while pairing them with independently verified strengths, helps us build credibility and guide customers towards the platform that best fits each workload and compliance profile.

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How to Discuss OCI Performance without Hype

1. Lead with outcomes, not components

Begin by asking what the workload must achieve: millisecond response for an online service, hour-ahead accuracy for an operational forecast, or predictable month-end close for finance. Link each requirement to a measurable performance target: latency, throughput, time-to-train, rather than to a specific OCI feature.

2. Use diagnostic questions to uncover constraints

- What variability in response time is acceptable before an SLA breach?
- How long does a full analytic cycle or model-training run last today?
- Which cost element (compute, storage, data egress) dominates the budget?
- Are there regulatory limits on where data may transit or reside?

Answers reveal whether off-box virtualisation, RDMA networking or low-cost egress will deliver tangible benefit.

3. Position OCI through reproducible evidence

Reference independent benchmarks (MLPerf, SPEC, TPC) and named customer results rather than authored metrics. Offer to rerun a trimmed-down version of the workload during a proof-of-concept so the client can validate performance personally.

4. Provide balanced comparisons

Acknowledge that hyperscalers offer broader regional coverage or a larger marketplace, then explain how OCI addresses the specific performance gap under discussion. Use phrases such as "AWS offers more functions in this category; OCI provides lower tail latency for tightly-coupled clusters." This framing shows respect for competitor strengths while clarifying OCI's differentiator.

Key take-aways for OCI ACE

- **Understand** how off-box virtualisation and an RDMA fabric remove host-CPU bottlenecks and deliver low-jitter performance for AI, HPC and Exadata workloads.
- **Position** OCI's price–performance advantage by referencing MLPerf, SPEC, TPC and independent cost studies rather than Oracle-authored benchmarks.
- **Demonstrate** customer outcomes such as NHS Shared Business Services' faster payments and National Grid ESO's shorter model-run times, to shift the conversation from lab tests to production impact.
- **Acknowledge** ecosystem size and regional-coverage gaps up front, and guide customers towards multi-cloud or partner solutions when those factors outweigh raw performance.
- **Offer** a transparent proof-of-concept using Terraform scripts and open monitoring so stakeholders can verify results under their own governance controls.
- **Link** OCI's low egress pricing and Azure Interconnect adjacency to budget oversight requirements common in UK public-sector projects.

Conclusion: performance by design, proven in the field

Oracle Cloud Infrastructure proves that price-performance is an engineering outcome, not a marketing promise. SmartNIC-based off-box virtualisation, a non-blocking 400 Gbps RDMA fabric, bare-metal tenancy and an all-NVMe storage tier eliminate the hidden taxes (hypervisor cycles, network oversubscription and spindle latency) that force other clouds to over-provision.

Independent evidence is unambiguous:

- **Performance:** OCI posts record-setting MLPerf GPT-3 times among public-cloud submissions, delivers sub-2 µs RDMA hop latency and reaches 1.3 M IOPS per compute instance on Block Volume.
- **Efficiency:** Independent SPECpower results still favour specialised on-prem servers, but an academic cross-cloud study finds OCI up to ~2.5 × cheaper per unit of compute performance than rival hyperscalers.
- **Production impact**: NHS Shared Business Services shortened supplier-payment batches; National Grid ESO halved model-run windows.

Those results satisfy procurement teams that demand an auditable trail because the underlying test scripts, tariff rates and runbooks are published for anyone to verify. Workloads still to move such as digital-forensics pipelines or national imaging archives, can validate the same gains through a scoped proof-of-concept under local governance controls.

By anchoring proposals in reproducible data and even acknowledging cases where another cloud may be a better fit, OCI ACE can guide public-sector organisations, with trust and integrity, to an infrastructure that meets mission-critical response times while respecting budgets, compliance obligations and public trust.